Lecture 12: Factors, curves and splines

- We have seen linear models as approximations of associations, using the simple linear regression.
- Liner models can do things beyond linearity!
- Categories can be used to predict an outcome too.
- Polynomials will be explored over simple straight curves. You see they used commonly. Not always well behaved.
- Splines (opens the door to other additive structures) will be introduced. They can be very flexible. But prone to over-fitting.

"Drawing" inference & Categories

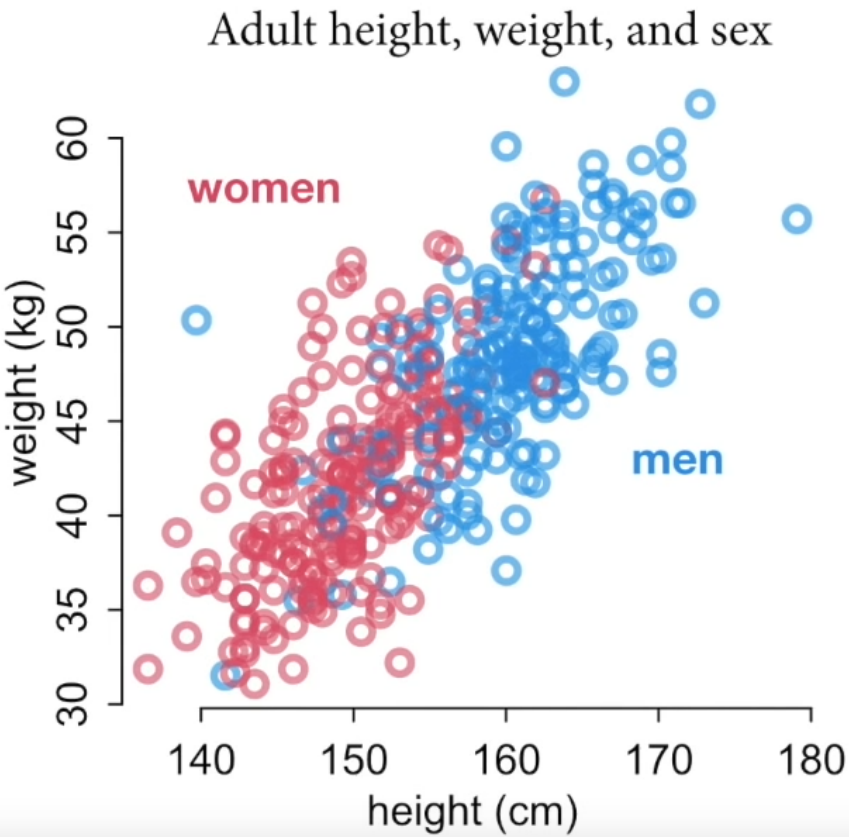
- How to use statistical models to get a scientific estimand.
- Need to incorporate causal thinking into:
  1. draw the statistical model and
  2. process the results.

- How to deal with non-continuous (discrete unordered) causes/predictors?
- Stratify by category: fit a separate line for each category. Sometimes OK.



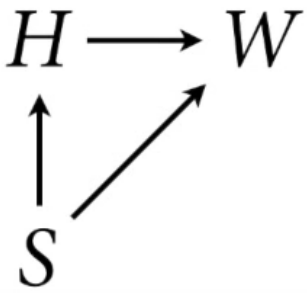
Howell weight vs height data with Sex added

```
> head(Howell1)
   height   weight age male
1 151.765 47.82561  63    1
2 139.700 36.48581  63    0
3 136.525 31.86484  65    0
4 156.845 53.04191  41    1
5 145.415 41.27687  51    0
6 163.830 62.99259  35    1
>
```



(we'll deal with Age later)

- To do something statistically, we need to think scientifically first.
- How are height, weight and sex related?



?

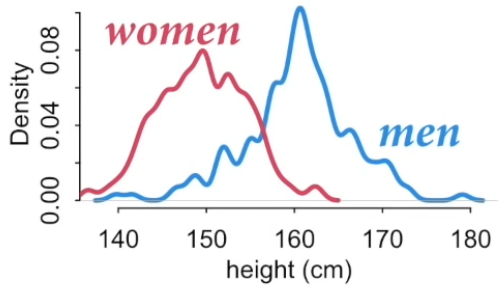
- We will build this DAG (directed acyclic graph in pieces)

The causes are not in the data

They are reflected in the data

$H \longrightarrow S$

$H \longleftarrow S$



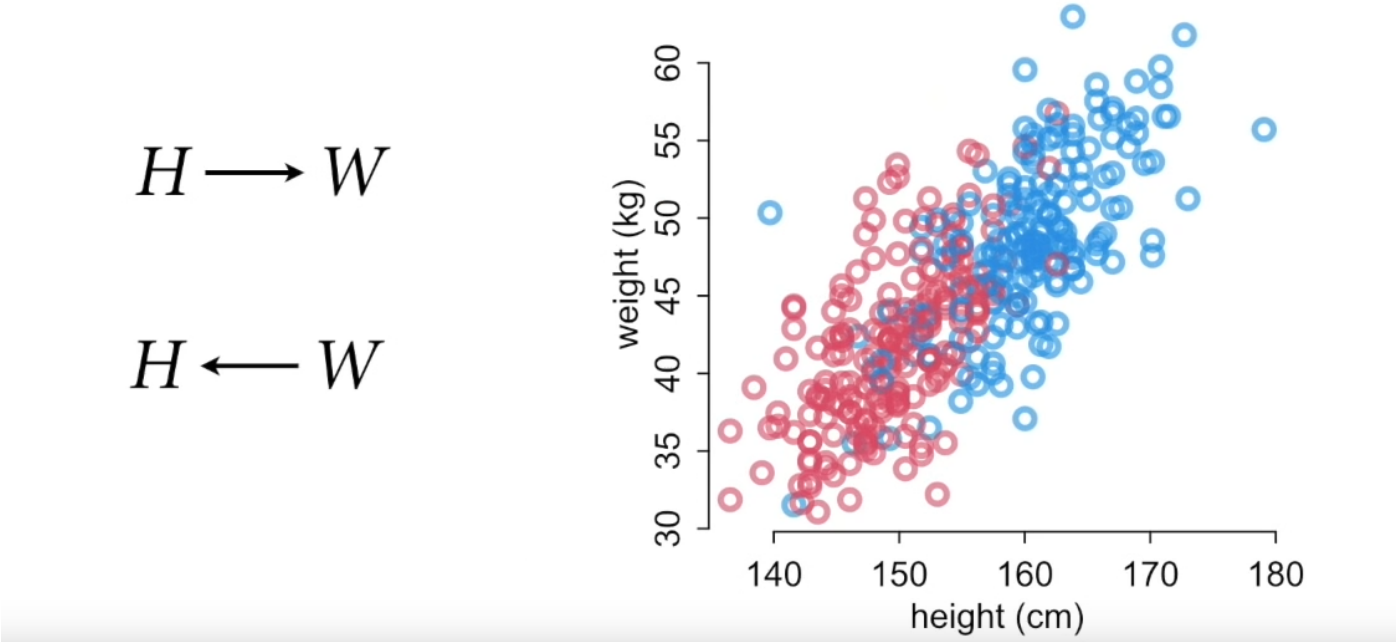
Both causal models can produce the same data. The former is more plausible.

The causes are not in the data

They are reflected in the data

$$H \longrightarrow W$$

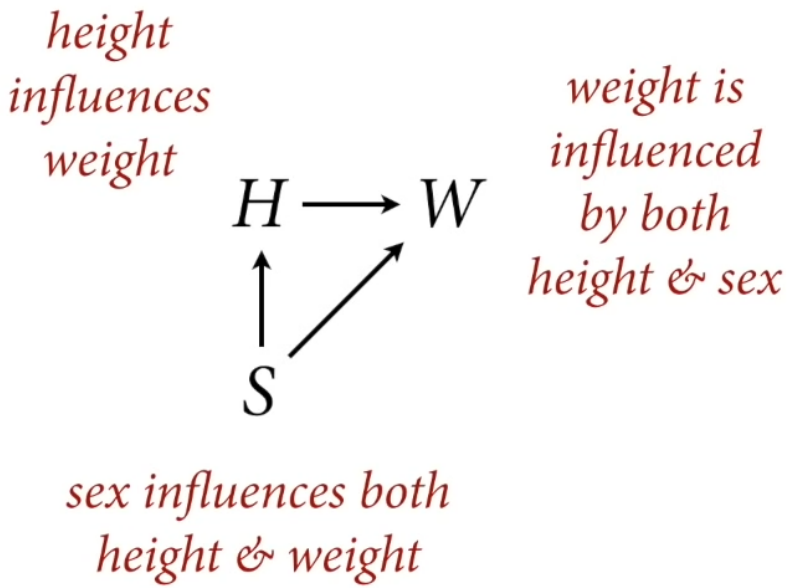
$$H \longleftarrow W$$



Both causal models can produce the same data. This time, the first one is a bit silly.
Similar trend for weight.
Use scientific background information.

Putting the pieces together

aka building the DAG



We need to specify the functions:

$$\begin{aligned} H &= f_H(S) \\ W &= f_W(H, S) \end{aligned}$$

Think scientifically first

From estimand to estimate

- Different causal questions need different statistical models:
  - Causal effect of H on W?
  - Causal effect of S on W? (H mediator)
  - Direct causal effect of S on W? (H is controlled for and path is blocked)
- For the last two questions, we need to include the variable S. As a categorical variable.
- Several ways to encore categorical variables: dummy/indicator variables (0/1) or index variables (1, 2, ..., k; order doesn't matter).
- We will use the latter (not what `lm()` does). It will allow us to extend modelling to more than two categories with no implementation change; it is also better to specify priors and it extend to mixed models without additional effort.

Modelling with categories

A first simple example

Say you want to model the effect α (1: cyan, 2: magenta, 3: yellow and 4: black) of using either of the four colours in your inkjet printer on the quality q of the drawing you print. You can use only one colour:

$$\begin{aligned} q_i &\sim \text{Normal}(\mu_i, \sigma) \\ \mu_i &= \alpha_{\text{COLOR}[i]} \end{aligned}$$

Now get back to the `Howell` data.

Using an index variable

Weight and Sex data

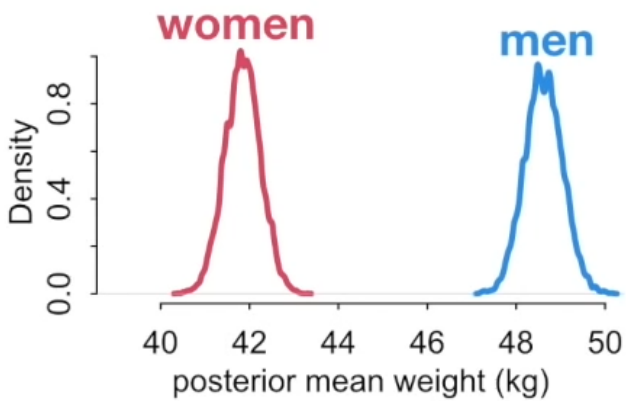
$$
\begin{aligned}
W_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha_{\text{Sex}[i]} \\
\alpha_j &\sim \text{Normal}(60, 10),\ j = 1, 2
\end{aligned}
$$

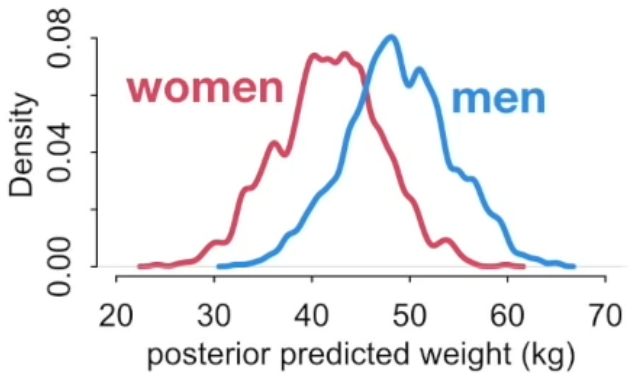
Two intercepts, one for each value of Sex, Sex=1 indicates female, Sex = 2 indicates male.
Priors, we keep the same for all categories. Neutral way to approach them. The code:

```
dat <- list(W=d2$weight, S=d2$male+1)
m_SW <- quap(alist(
  W ~ dnorm(mu, sigma),
  mu <- a[S],
  a[S] ~ dnorm(60, 10),
  sigma ~ dunif(0,10)
), data = dat)
```

Posterior mean and predictions

```
post <- extract.samples(m_SW, n = 10000)
dens(post$a[,1], xlim = c(39,50), lwd = 3, col = 2,
   xlab = "posterior mean weight (kg)")
dens(post$a[,2],lwd = 3, col = 4, add = TRUE)
```





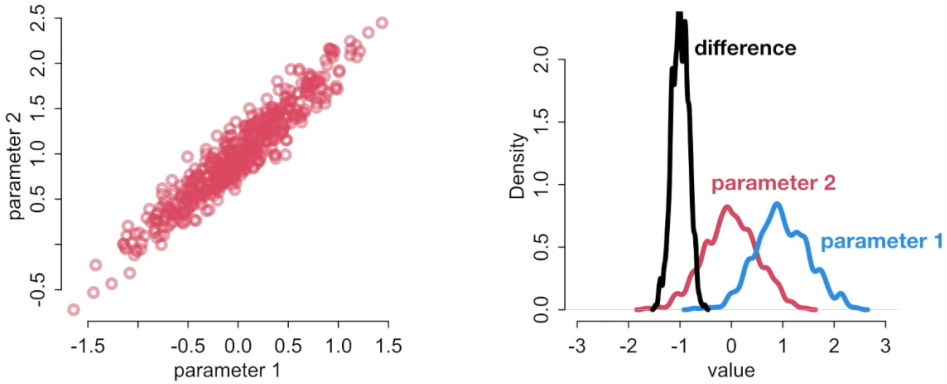

```
W <- rnorm(1000, post$a[,1], post$sigma)
M <- rnorm(1000, post$a[,2], post$sigma)
```

And you can plot these posterior predictions. Men are heavier on average, but are not necessarily heavier individually.

Contrasting

To make sense of the overlap

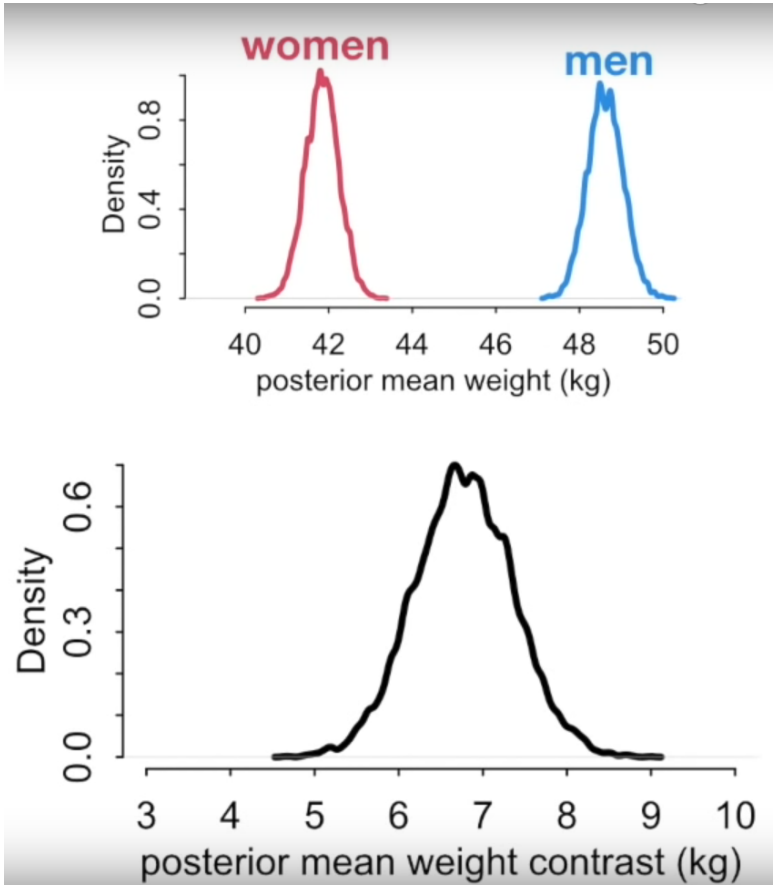
- Need to compute the *contrast*, the difference between categories.
- It is never legitimate to compare overlap between parameters. Because parameters are correlated with others.



- $\rightarrow$ Compute contrast distribution, i.e. the difference in predicted weights between M and F in this population.
- $\rightarrow$ What is the counterfactual effect if changing someone's sex?

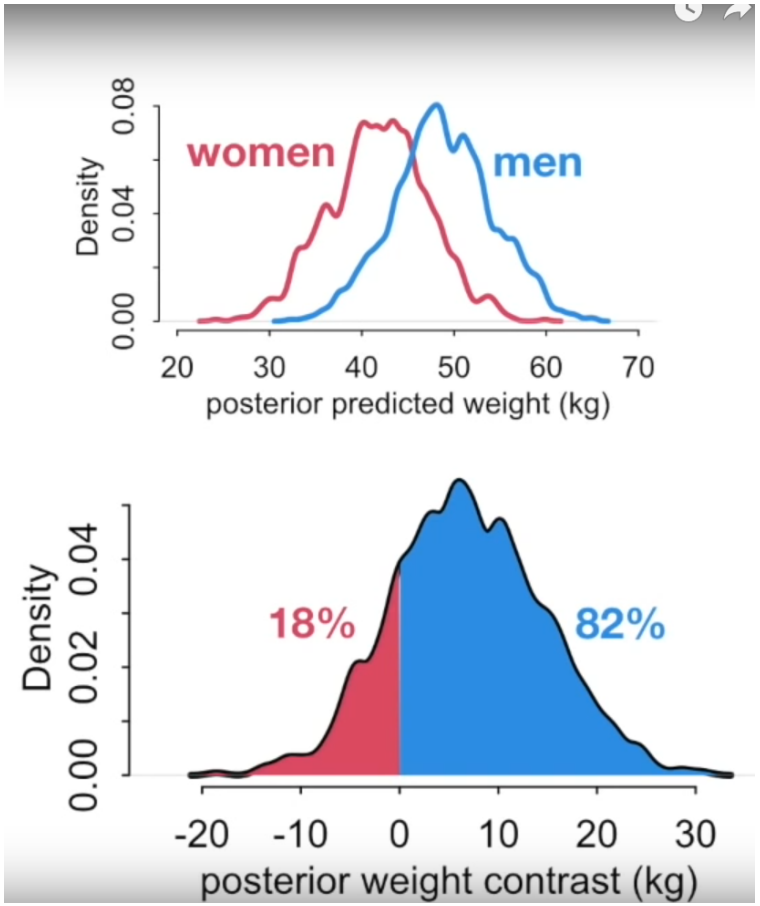
Causal (mean) contrast

```
# Causal contrast in means
mu_contrast <- post$a[,1]-
  post$a[,2]
dens(mu_contract, xlim = c(3,10)
  lwd = 3, col = 1, xlab =
  "posterior mean weight
  contract (kg)")
```



Weight contrast

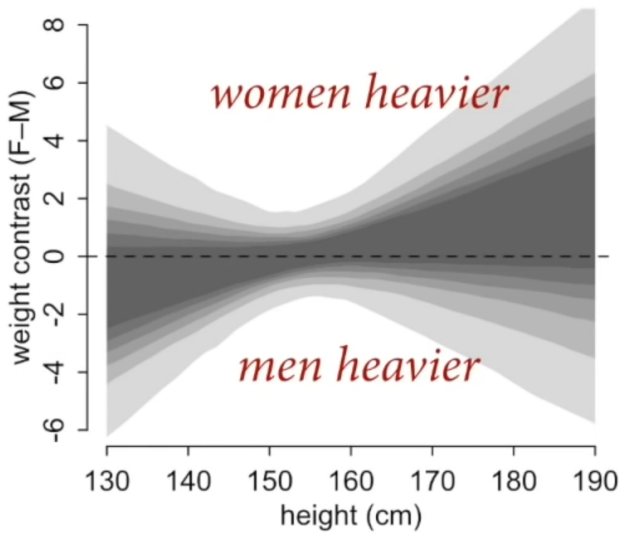
```
W.F <- rnorm(1000, post$a[,1],
  post$sigma)
W.M <- rnorm(1000, post$a[,2],
  post$sigma)
W.contrast <- W.M-W.F
dens(W.contrast, xlim = c(-25,
  35), lwd = 3, col = 1, xlab =
  "posterior weight contrast")
#Proportions above/below zero
sum(W.contrast>0)/1000
sum(W.contrast<0)/1000
```



Now weight as a function of height & sex

$$
\begin{aligned}
W_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \alpha_{\text{Sex}[i]} + \beta_{\text{Sex}[i]}(H_i - \bar{H})
\end{aligned}
$$

Do the coding (only need to add `beta[S]` to the definition of `mu` and a prior for it; see in a few slides too) to get:



For F and M of same height, if they are short (take 140cm), M tend to be heavier. If they are tall (take 180cm), men tend to be lighter.

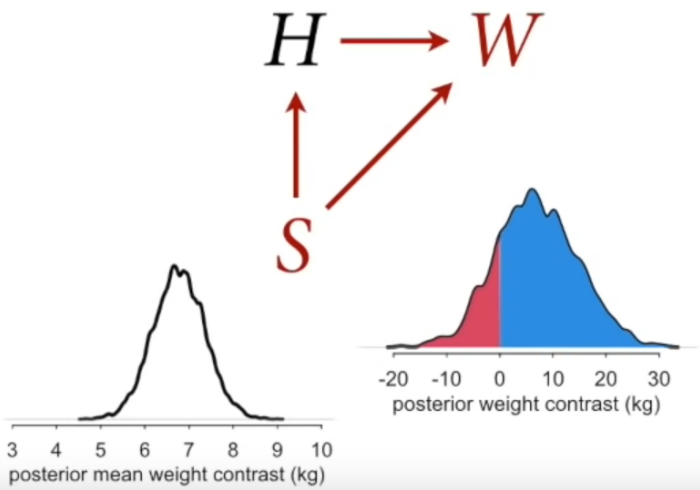
Nearly all the causal effect of S on W acts through H.

And we want to (i) compute two predictive posterior distributions for F and M, respectively (ii) do the contrast and (iii) plot the distribution of the difference at each height.

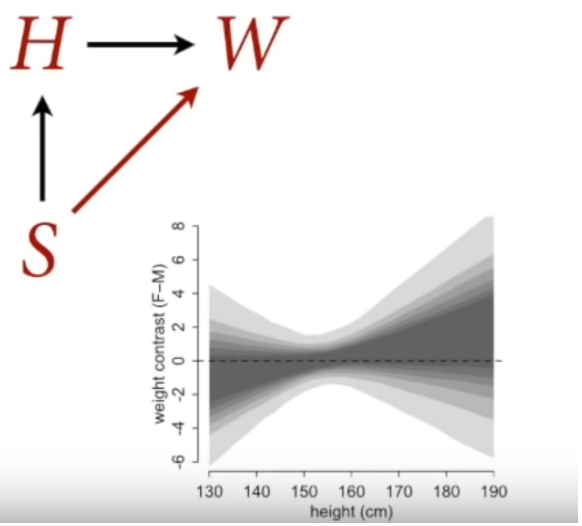
From estimand to estimate

Bayes luxury

Q: Causal effect of S on W?



Q: Direct causal effect of S on W?



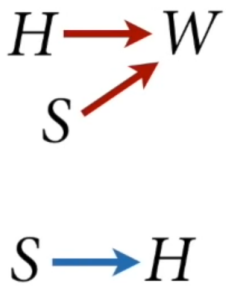
We used 2 models for 2 estimands. Alternative approach is to use *ONE* model for the entire causal system. Then use joint posterior to compute the desired estimandS.

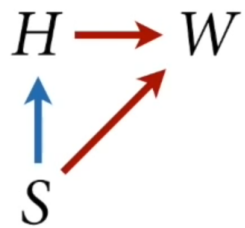
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One Bayes model to rule them all

Running the two models simultaneously:

```
m_SHW_full <- quap(alist(
  W ~ dnorm(mu, sigma),
  mu <- a[S]+b[S]*(H-Hbar),
  a[S] ~ dnorm(60, 10),
  b[S] ~ dlnorm(0,1),
  sigma ~ dunif(0,10),
  H ~ dnorm(nu, tau),
  nu <- h[S],
  tau ~ dunif(0,10)), data = dat)
```





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Simulating interventions

To assess causal effect

Total causal effect of S on W, aka simulating "changing the sex at birth"

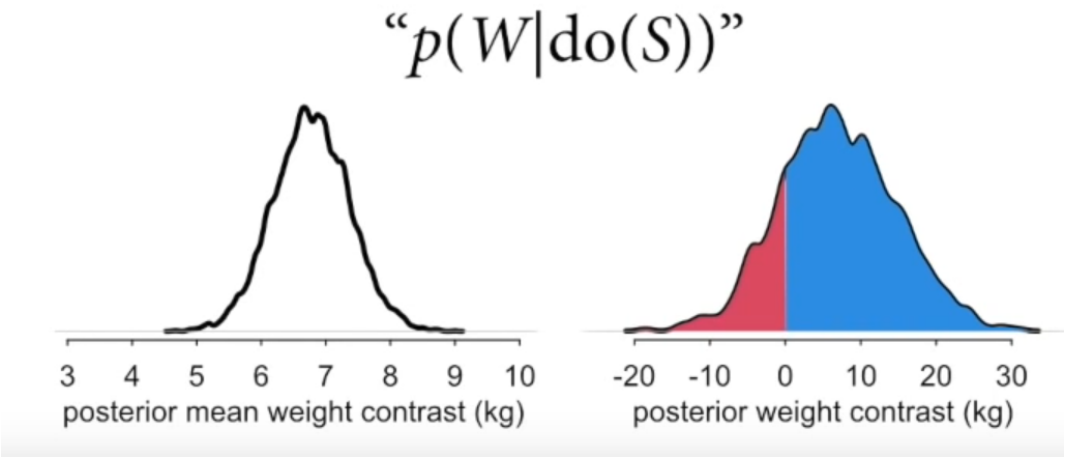
```
post <- extract.samples(m_SHW_full)
hbar <- dat$Hbar
n <- 1e4

with(post, {
# Simulate W for S=1 (F)
H_SF <- rnorm(n, h[,1], tau)
W_SF <- rnorm(n, a[,1] + b[, 1]*(H_SF-hbar), sigma)
# Simulate W for S=2 (M)
H_SM <- rnorm(n, h[,2], tau)
W_SM <- rnorm(n, a[,2] + b[, 2]*(H_SM-hbar), sigma)
# Compute contrast
W_do_S <<- W_SM-W_SF})
```

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Total causal effect of S on W

Simuated consequence of changing S at birth



Automated way to do it:

```
HW_sim <- sim(m_SHW_full, data = list(S=c(1,2),
  vars = c("H", "W"))
  W_do_Sauto <- HWsim$W[,2]-HWsim$W[,1]
```

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Categorical variables

Take-home message

- Easy to use with index coding.
- Must later use samples to compute relevant contrasts.
- Always summarise at the last step.
- Want mean difference and NOT difference in means $\rightarrow$ make sure you account for the variability in the posterior parameters or posterior predictions
- We saw we could (i) design a unique model for each estimand and compute it or (ii) compute a joint posterior from the full causal system and simulate each estimand via interventions.

- Not sure how long we still have this week...

Polynomial regression

- Purely descriptive strategy
- Using polynomials or descriptive variable:

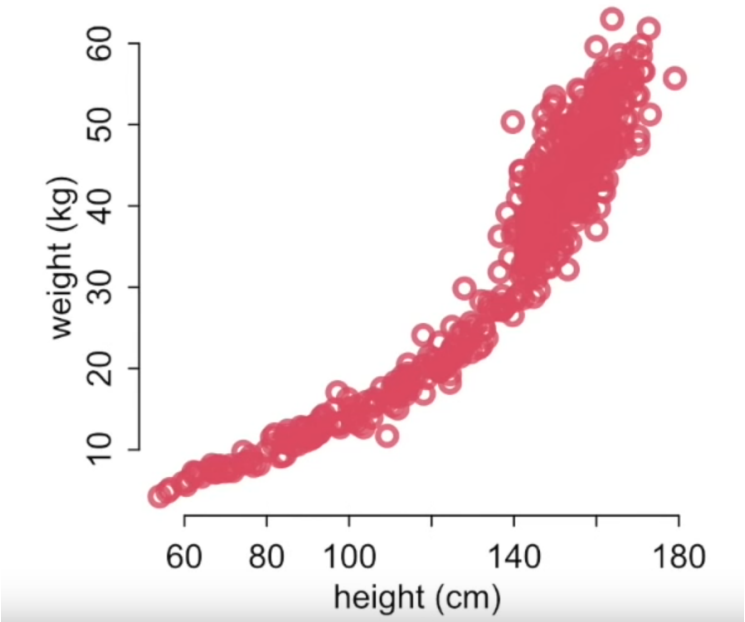
$$
\begin{aligned}
\mu_i &= \alpha + \beta_1 x_i \\
\mu_i &= \alpha + \beta_1 x_i + \beta_2 x_i^2 \\
&\dots \\
\mu_i &= \alpha + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_k x_i^k
\end{aligned}
$$

- Second order is the equation of a parabola.
- The more terms, the more flexible.
- Problems: strange symmetries and explosive uncertainty at the boundaries.
- See `lecture12-parabola.mp4` video.

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Polynomial regression

Using the full Kung! weight/height data



- $H \rightarrow W$ not linear.
- Linear models can fit this curve.
- But this is not mechanistic.

Standardised predictors

- Very helpful to standardise predictors before fitting
  - Makes estimation easier.
  - Can make the interpretation easier.
- To standardise:
  - Subtract the mean.
  - Divide by standard deviation.
  - The predictor has mean zero and sd of 1.

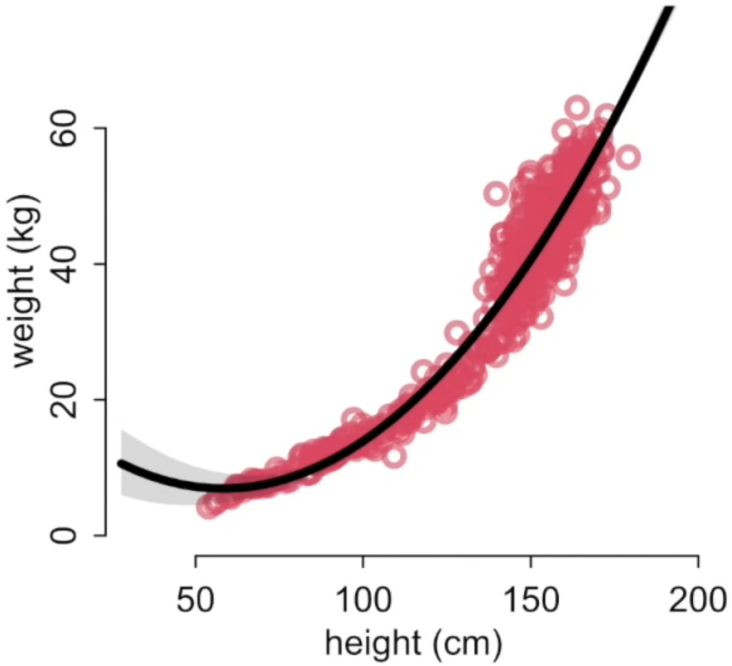
Parabolic model of weight

```
d$height_s <- (d$height - mean(d$height))/sd(d$height)
d$height_s2 <- (d$height_s)^2
m4.5inv <- quap(alist(
 weight ~ dnorm(mu, sigma),
 mu <- alpha+b1*height_s+b2*height_s2,
 a <- dnorm(178, 20),
 b1 ~ dlnorm(0,1),
 b2 ~ dnorm(0,1),
 sigma ~ dunif(0,50)
 ), data = d
)
```

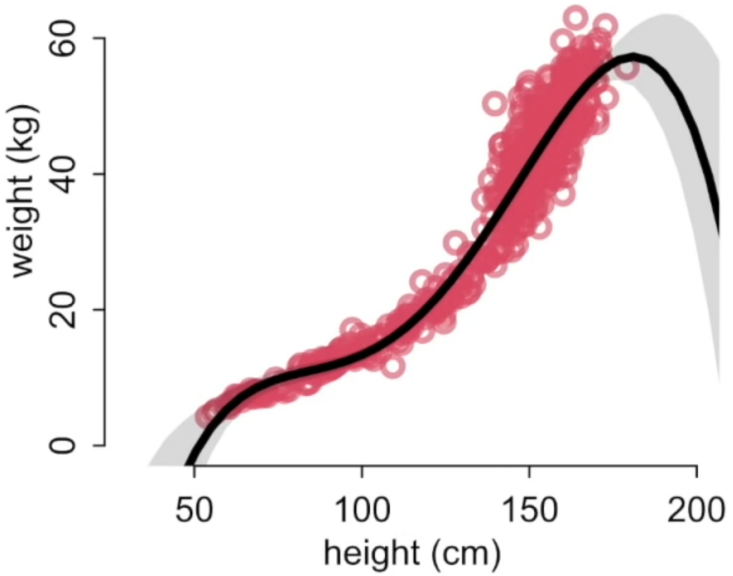
Fitting a parabola or a polynomial of order 4

to the Kung! Weight vs Height data

$$\mu_i = \alpha + \beta_1 H_i + \beta_2 H_i^2$$



$$\mu_i = \alpha + \beta_1 H_i + \beta_2 H_i^2 + b_3 H^3 + b_4 H^4$$



Not that flexible, cannot get a monotonic curve for example.

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From local to global - B-splines

- Basis-splines: wiggly functions built from many local less wiggly functions.
- Basis function: a local function. An elementary building block translated and shrunk/expanded.
- Better than polynomials but still very geocentric.
- Bayesian B-splines often coined P-splines.
- Essentially linear models with some weird synthetic variables

$$\mu_i = \alpha + w_1 \times B_{i,1} + w_2 \times B_{i,2} + w_3 \times B_{i,3} + ...$$

Weights are like slopes and B local functions (not data-dependent) turn on-off (dimmer-switch) the weights in different regions.

What a real-world spline is



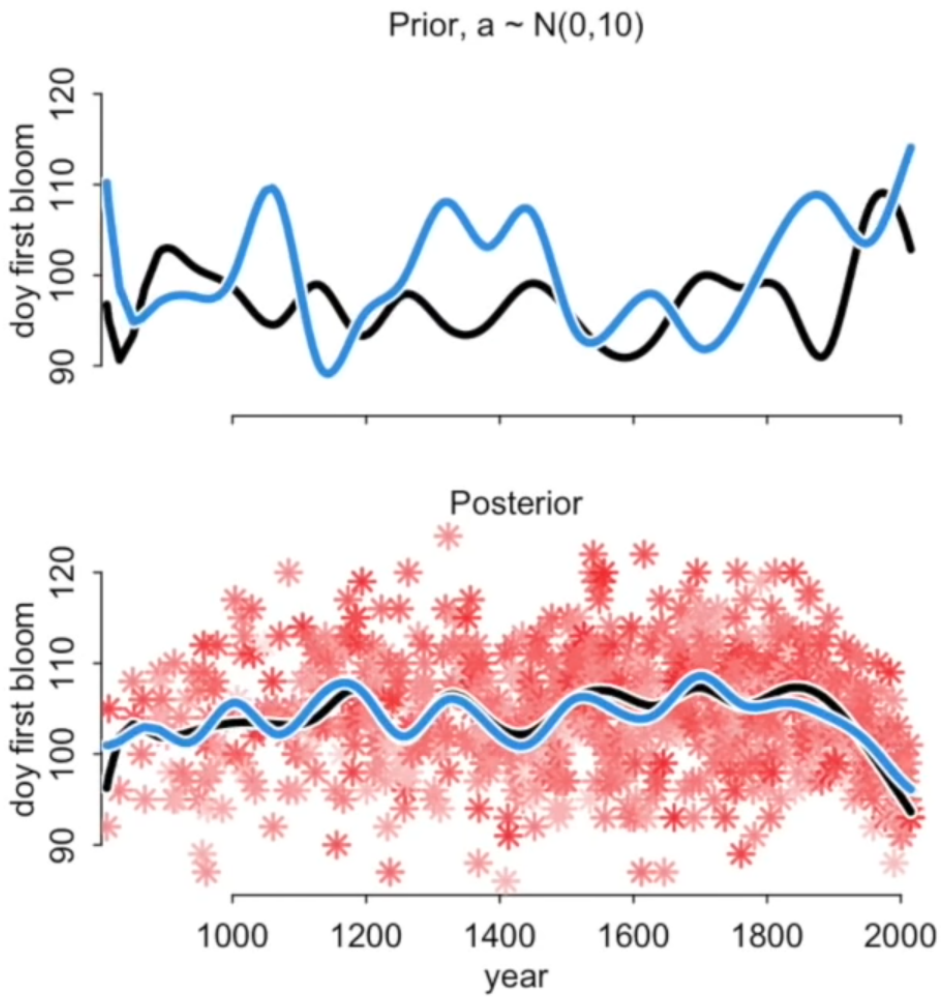
Cherry blossom data

```
library(rethinking)
data(cherry_blossom)
d <- cherry_blossom
precis(d)
```



Splines

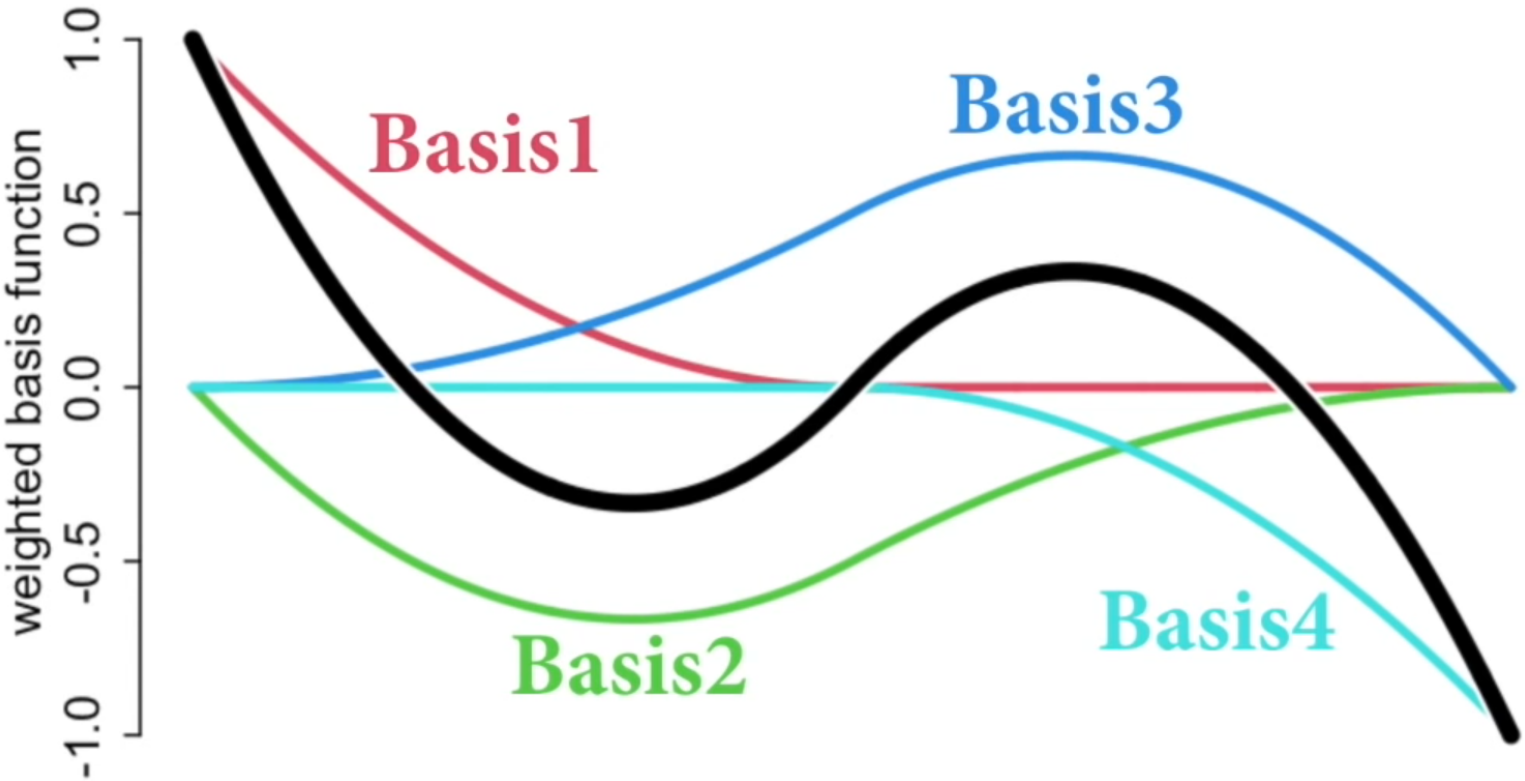
From prior flexibility to fitting the data, posterior predictive distribution



B-splines recipe

- Linear regression on synthetic variables (as many as you want them)
- Divide the predictor variable range into regions thanks to *knots*, i.e. spline anchors.
- Choose degree (complexity) of basis function: how wiggly?
- Note also: more knots = more wiggle (aka less smoothness) in the global spline function.
- Find posterior distribution of weights.

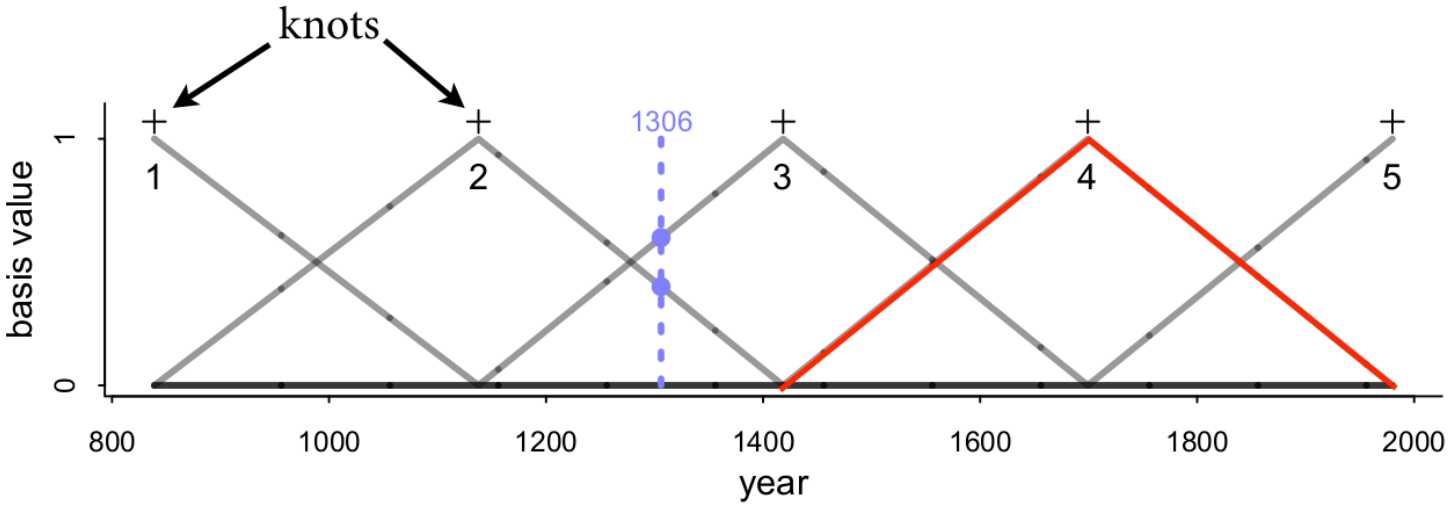
From basis function to the spline



$w = [1, -1, 1, -1]$

Starter example

linear basis functions



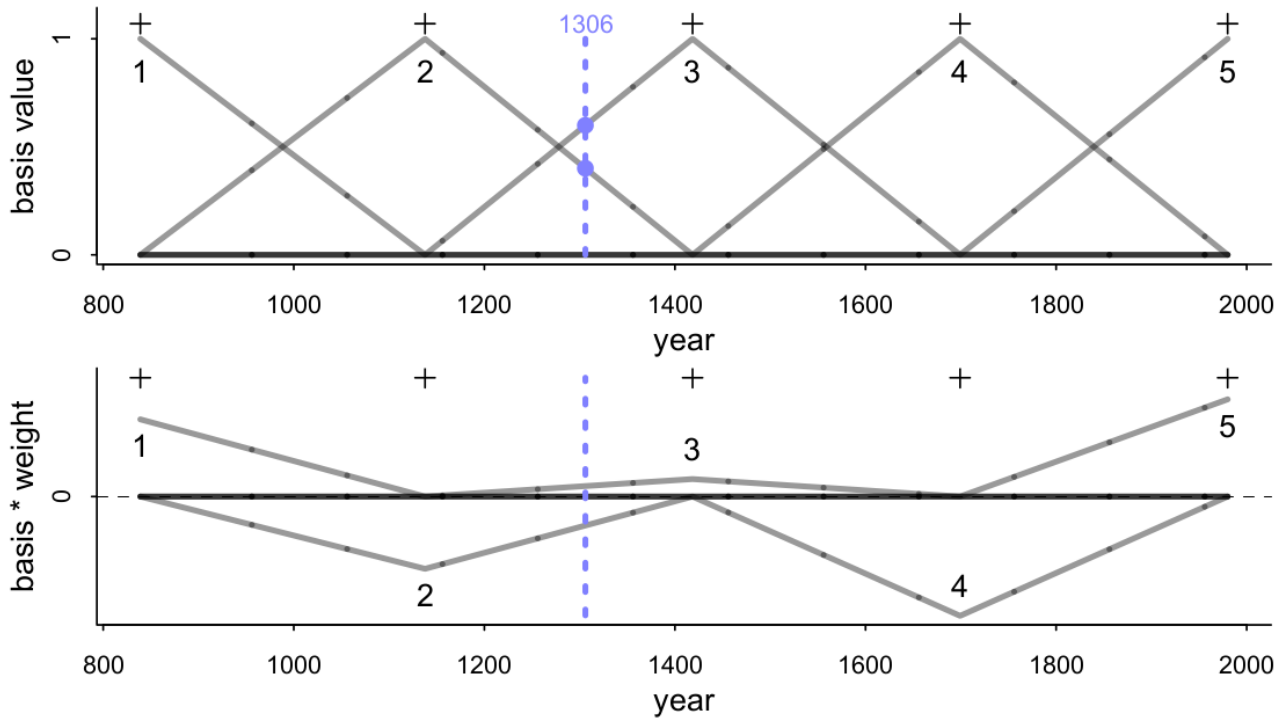
Each basis defines the local region, where it influences the spline.

Fitting the weights

Yet another ordinary linear model

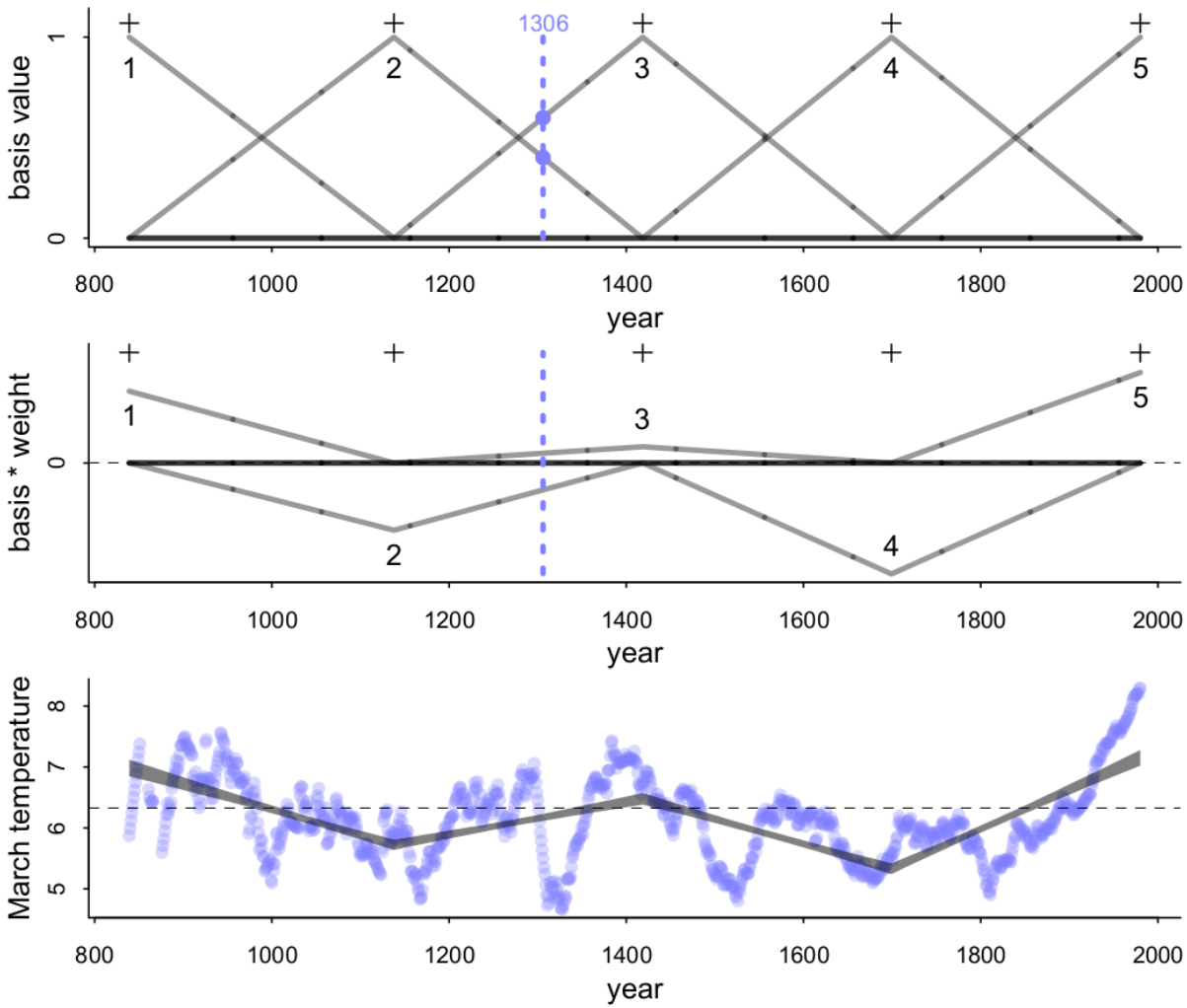
```
m4.7 <- quap(alist(
   T ~ dnorm(mu. sigma)
   mu <- a + B %*% w
   a ~ dnorm(6, 10)
   w ~ dnorm(0,1)
   sigma ~ dexp(1)),
   data = list(T = d$temp, B=B),
   start = list(w=rep(0, ncol(B)))
)
```

Adding (rather multiplying) the weights



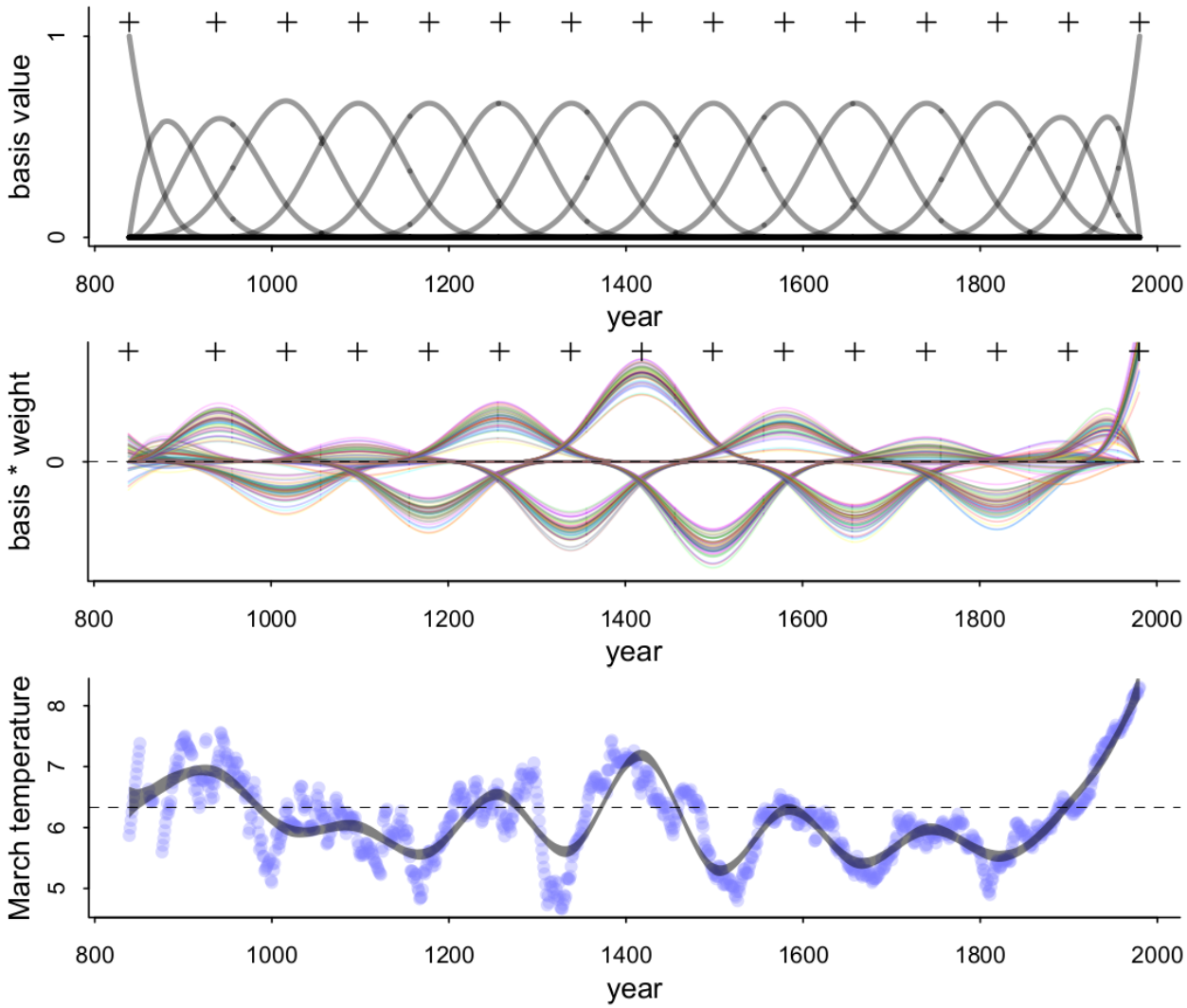
And plotting the spline

next to the data



Spline fitting

15 knots, 3rd degree splines



(all code in the SR book)

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The world of splines

- Knots and basis functions are choices.
- Must worry about overfitting the data (see later in the course)
- Some types of splines don't require knots.
- Another idea: Gaussian processes (see Chapter 14 in SR book; not in this course).
- All splines are descriptive, NOT mechanistic, I repeat.

Watch second lecture video `lecture12-splines.mp4`.

Curves and splines

- Can be very non-linear functions from linear pieces.
- Polynomials and splines are very powerful geocentric models.
- Adding scientific information always helps. E.g. Weight only increases with Height or Height only increases with Age and then levels off (or declines).
- Ideally, statistical models have some form of scientific models.

